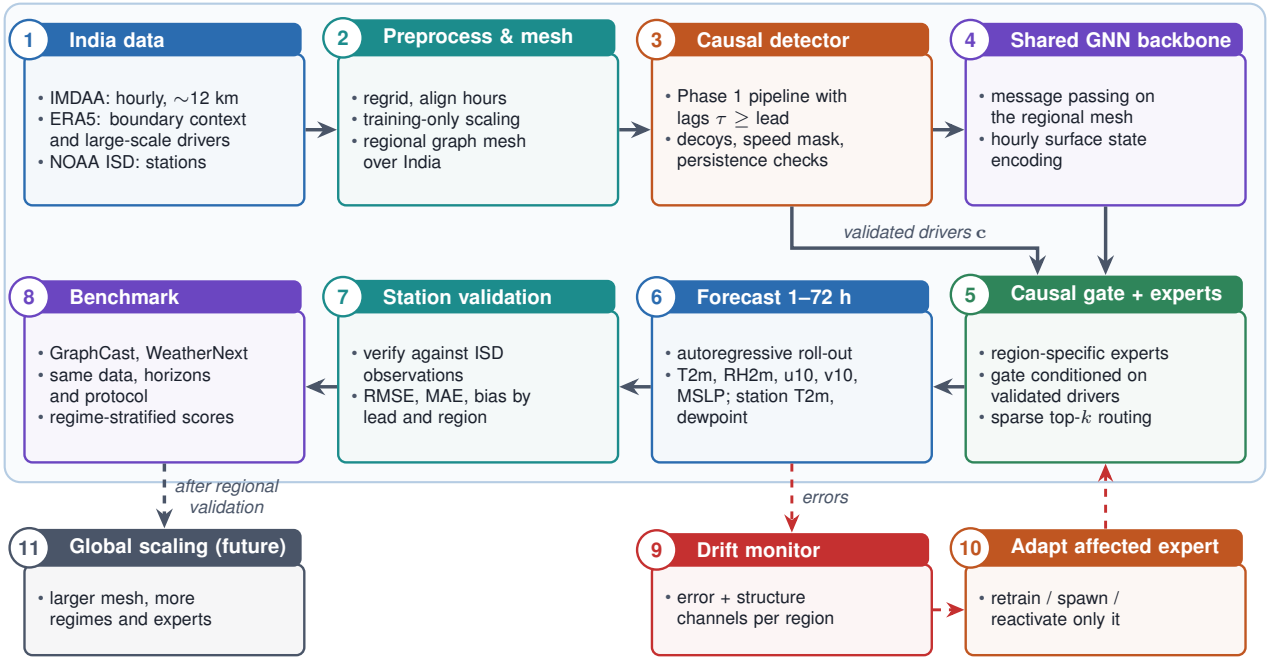

Phase 2: India-focused hourly forecasting with causally gated experts
1
India data
• IMDAA: hourly, ∼12 km
• ERA5: boundary context and large-scale drivers
• NOAA ISD: stations
2
Preprocess & mesh
• regrid, align hours
• training-only scaling
• regional graph mesh over India
3
Causal detector
• Phase 1 pipeline with lags $\tau \geq$ lead
• decoys, speed mask, persistence checks
4
Shared GNN backbone
• message passing on the regional mesh
• hourly surface state encoding
validated drivers $\mathbf{c}$
5
Causal gate + experts
• region-specific experts
• gate conditioned on validated drivers
• sparse top-k routing
6
Forecast 1–72 h
• autoregressive roll-out
• T2m, RH2m, u10, v10, MSLP; station T2m, dewpoint
7
Station validation
• verify against ISD observations
• RMSE, MAE, bias by lead and region
8
Benchmark
• GraphCast, WeatherNext
• same data, horizons and protocol
• regime-stratified scores
errors
9
Drift monitor
• error + structure channels per region
10
Adapt affected expert
• retrain / spawn / reactivate only it
after regional validation
11
Global scaling (future)
• larger mesh, more regimes and experts